ONLINE ASSESSMENT · 8 OCT 2026

Pace Stock

Roles: Software Engineer · Jr. Quant Researcher

1. Questions

Pattern: 4 coding questions · 90 minutes

Q1. Shortest Walk Through Required Vertices Coding

A tree has n nodes $1 \dots n$ and $n-1$ edges (each costs 1 to walk). You are given k vertices $a_1, \dots, a_k$ that must be visited.

- The walk **starts at node 1** and must **end at node** n.
- Every a_i must be visited at least once, in any order. Nodes and edges may be repeated.

Return: the minimum number of edges walked.

> Practice (same problem, any start/end): Codeforces 1675F — Vlad and Unfinished Business

Q2. Nearest City on the Same Line Coding

There are n cities: `city[i]` is at (x_i, y_i), given by `city` (`vector<string>`), `x`, `y` (`vector<int>`). No two cities are at the same point. You are given q queries, each a city name.

- For a queried city, the candidates are the **other** cities with the **same** x or the **same** y.
- Answer = the nearest candidate. If two are equally near, the lexicographically smaller name.
- If no other city shares its x or y, the answer is `"NONE"`.

Return: `vector<string>` with the answer for every query, in order.

Q3. Exactly Overlapping Regions Coding

You are given two binary grids `grid1` and `grid2` of the same size $R \times C$. A **region** is a group of 1-cells connected 4-directionally (up, down, left, right).

- Place `grid2` on top of `grid1`. A region of `grid2` counts if `grid1` has a region made of **exactly the same cells** (no cell more, no cell less).

Return: the number of such regions of `grid2`.

> Practice (closest, weaker "inside" condition): LeetCode 1905 — Count Sub Islands

Q4. Suffix-to-Front Transformations Coding

You are given strings s and t of equal length n, and an integer k.

- One **transformation**: remove a suffix of length l ($0 < l < n$) and append it at the front. E.g. `abcd` with $l = 2$ → `cdab`.
- Perform **exactly** k transformations.

Return: the number of ways to turn s into t, modulo $10^9 + 7$.

> Practice (same problem): LeetCode 2851 — String Transformation